Chapter 2 Answers

March 30, 2024

Lemma

We must show that, from F_1 and F_2, the following two conditions hold:
(1): We need to show that $\sim\exists x\phi(x)$ is true iff $\sim\phi(a)$ is true for every parameter a. Suppose that $\sim\exists x\phi(x)$ is true (under some valuation v). Thus, by F_2, the sentence $\phi(a)$ is not true under v for any parameter a. Because $\phi(a)$ is not true under v for any parameter a, it follows that $\sim\phi(a)$ is true under v for every parameter a (by B_1). On the other hand, suppose that $\sim\phi(a)$ is true under v for every parameter a. Thus, by B_1, $\phi(a)$ is false under v for every parameter a (by B_1). Thus, by F_2, $\exists x\phi(x)$ is not true under v, meaning, by B_1, $\sim\exists x\phi(x)$ is true under v.
(2): We need to show that $\sim\forall x\phi(x)$ is true iff $\sim\phi(a)$ is true for at least one parameter a. On the one hand, suppose that $\sim\forall x\phi(x)$ is true under some valuation v. Thus $\forall x\phi(x)$ is not true under v (by B_1), meaning, by F_1, $\phi(a)$ is not true for every parameter a. Thus, for the parameter e for which $\phi(e)$ is not true, $\sim\phi(e)$ is true by B_1. On the other hand, suppose that $\sim\phi(a)$ is true for at least one parameter a, call it e. Thus, by B_1, $\phi(e)$ is false, meaning, by F_1, $\forall x\phi(x)$ is not true, and therefore, by B_1, $\sim\forall x\phi(x)$ is true.

1

We suppose that v is a Boolean valuation such that the following two conditions hold (for all sentences γ and δ):
(1) if γ is true under v, then for every parameter a, the sentence $\gamma(a)$ is true under v, and
(2) if δ is true under v, then $\delta(a)$ is true under v for at least one parameter a.
We must prove that v is a first-order valuation. Thus, we must derive F_1' and F_2'. Because we have already have (1) and (2), we only need to prove the converses

of (1) and (2).
(1): In this case, we need to prove that, if for every parameter a, the sentence $\gamma(a)$ is true under v, then γ is true under v.

Case 1: In this case, γ is of the form $\forall x\phi(x)$. So suppose that $\phi(a)$ is true under v for every paramter a. By B_1, it follows that $\sim\phi(a)$ is false under v for every parameter a, meaning, by (2), that $\sim\forall x\phi(x)$ is not true, and is therefore false (because every sentence X is either assigned the value t or f), meaning, by B_1, $\forall x\phi(x)$ is true.

Case 2: In this case, γ is of the form $\sim\exists x\phi(x)$. So suppose that $\sim\phi(a)$ is true for every parameter a, meaning, by B_1, that $\phi(a)$ is false for every parameter a. Thus, by (2), $\exists x\phi(x)$ is not true, and is therefore false (for the same reason as in Case 1), meaning, by B_1, that $\sim\exists x\phi(x)$ is true.

(2): In this case, we need to prove that, if $\delta(a)$ is true for at least one parameter a under v, then δ is true under v.

Case 1: In this case, δ is of the form $\exists x\phi(x)$. So suppose that $\phi(a)$ is true for at least one parameter a, call it e. By B_1, $\sim\phi(e)$ is false, meaning, by (1), $\sim\exists x\phi(x)$ is not true, and is therefore false, and thus, by B_1, $\exists x\phi(x)$ is true.

Case 2: In this case, δ is of the form $\sim\forall x\phi(x)$. So suppose that $\sim\phi(a)$ is true for at least one parameter a, call it e. By B_1, $\phi(e)$ is false, meaning, by (1), $\forall x\phi(x)$ is not true, and therefore false, meaning, by B_1, $\sim\forall x\phi(x)$ is true.

2

We are to show that every Magic set is a truth-functional basis for propositional logic.

On the one hand, suppose that some pure sentence X is valid. Thus, it is true under all first-order valuations. We must show that X is a tautological consequence of some finite subset M_0 of M. That is, we must show that under every *Boolean* valuation that satisfies M_0, X is true. Well, by M_1, every Boolean valuation that satisfies S is also a first-order valuation. Thus, X is true under all Boolean valuations that satisfy M, because X was assumed to be true under all first-order valuations, and every Boolean valuation that satisfies M is also a first-order valuation. Thus, X is a tautological consequence of M. By the Corollary to the Denumerable Compactness Theorem for Propositional Logic, it is also a logical consequence of some finite subset M_0 of M.

On the other hand, suppose that some pure sentence X is a truth-functional consequence of some finite subset M_0 of M. Thus, $M_0 \bigcup \sim X$ is not first-order satisfiable (every first-order valuation is also a Boolean valuation). This means that $\sim X$ is not first-order satisfiable. For, if it were, by M_2, $M_0 \bigcup \sim X$ would also be first-order satisfiable. Thus, X is valid (true under every first-order valuation).

3

We must show that every complete regular set is also a magic set. Recall that a complete regular set M is a set such that for every γ, $\gamma \supset \gamma(a) \in M$ for every parameter a, and for every δ, $\delta \supset \delta(a) \in M$ for at least one parameter a, and that every finite subset of M is a regular set. Because we are showing that every complete regular set is a magic set, it suffices to show that M satisfies the conditions M_1, M_2.

M_1: We must show that every Boolean valuation that satisfies M is also a first-order valuation. To that end, it suffices to show that:
If γ is true, then $\gamma(a)$ is true for every parameter a,
If δ is true, then $\delta(a)$ is true for at least one parameter a.
(which implies a first-order valuation by Problem 1). So, suppose γ is true. Thus, $\gamma \supset \gamma(a) \in M$ for every parameter a. Because this is a Boolean valuation, and γ is true, $\gamma(a)$ is likewise true for $\gamma(a)$ for every parameter a (by B_4). On the other hand, suppose that δ is true. Thus, $\delta \supset \delta(a) \in M$ for at least one parameter a. Again, because this is a Boolean valuation, and δ is true, $\delta(a)$ is likewise true for at least one parameter a (again by B_4).

M_2: We must show that, for every finite set S_0 of pure sentences, and every finite subset M_0 of M, if S_0 is first-order satisfiable, so is $M_0 \bigcup S_0$. So, suppose that S_0 is a finite set of pure sentences, and that it is first-order satisfiable. Now consider any finite subset M_0 of M. By the definition of a complete regular set, it is a regular set. Thus, it can be split into a sequence

$$Q_1 \supset Q_1(a_1), ..., Q_n \supset Q_n(a_n)$$

where, if Q_i is a γ, a_i is any parameter a, and if Q_i is a δ, then a_i is a parameter a such that a does not occur in δ or in any previous Q's. So, what we really need to show is that, if a set S is first-order satisfiable, $S \bigcup \gamma \supset \gamma(a)$ is first-order satisfiable for every parameter a, and $S \bigcup \delta \supset \delta(a)$ is first-order satisfiable, provided a does not occur in δ or S. This would let us show that

$$S_0 \bigcup Q_1 \supset Q_1(a_1) \bigcup ... \bigcup Q_n \supset Q_n(a_n)$$

is first-order satisfiable, which is the same as saying $S_0 \bigcup M_0$ is first-order satisfiable. There are then two cases: Q_i is a γ, or is a δ.

γ: in this case, Q_i is a γ, meaning $Q_i \supset Q_i(a_i)$ is of the form $\gamma \supset \gamma(a_i)$. Suppose S is first-order satisfiable under some interpretation I. Well, $\gamma \supset \gamma(a_i)$ is satsified by *every* first-order valuation (which can easily be shown, if one desires), meaning $S \bigcup \gamma \supset \gamma(a_i)$ is obviously also first-order satisfiable.

δ: in this case, Q_i is a δ, meaning $Q_i \supset Q_i(a_i)$ is of the form $\delta \supset \delta(a_i)$. Suppose S is first-order satisfiable under some interpreation I. If δ is false under I, then $\delta \supset \delta(a_i)$ is true under I. If δ is true under I, then $S \bigcup \delta$ is satisfiable, meaning $S \bigcup \delta \supset \delta(a_i)$ is satisfiable by Problem 2 of Chapter 9 of the Beginners Guide[Smullyan, 2014] (because we assumed that a does not occur in δ or any of the previous $Q_i's$, and that S_0 was a set of pure sentences, meaning a of course does not occur in them). In either case, $S \bigcup \delta \supset \delta(a_i)$ is satsifiable.

4

We are to find a quicker proof of the Regularity theorem, using Magic sets. It has been shown that a complete regular set exists, and by Problem 3, it is a Magic set. By Problem 2, it is therefore a truth-functional basis for first-order logic. So, suppose that X is a valid pure sentence. Thus, X is truth-functionally implied by some finite subset M_0 of M. By the definition of a complete regular set, M_0 is a regular set. Thus, X is tautologically implied by some regular set, which is what we were required to prove.

System G_0

Correctness of G_0

The Axiom of the system is obviouslly correct. For, whenever the set of formulas U, X is true, X is true, meaning it is obviously true that *at least one* of the formulas V, X are true for any set of formulas V.

C_1: This is obvious, as X, Y are defined to both be true whenever $X \wedge Y$ is true.

C_2: If whenever all elements of U are true, at least one element of V, X and V, Y is true, on the one hand, if V is the element which is true, it follows that at least one element (V) of $V, X \wedge Y$ is true. On the other hand, if V is false, it would then follow that both X, Y are true, therefore meaning $X \wedge Y$ is the element which is true.

D_1: If whenever all elements of U are true then at least one element of V, X, Y is true, if V is the element which is true, it follows that at least one element of (V) of $V, X \vee Y$ is true. On the other hand, if V is false, it would then follow that either X or Y is true, therefore meaning at least one element of $V, X \vee Y$ is true.

D_2: We are given that at least one element of V is true whenever both U, X are true, or when both U, Y are true. By definition of $X \vee Y$, it follows that all elements of $U, X \vee Y$ are true iff either X or Y is true, which would imply that at least one element of V is true.

I_1: We are given that at least one element of V, Y is true whenever all elements of U, X are true. Thus, if all elements of U are true, either X is true or false. If it is false, at least one element $(X \supset Y)$ of $V, X \supset Y$ is true, while if it is true, it would follow by the assumptions that either V or Y is true. If it is V that is true, this is the element that is true. If it is Y that is true, then $X \supset Y$ is also true.

I_2: We are given that at least one element of V, X is true whenever U is true, and that at least one element of V is true whenever all elements of U, Y are true. Thus, if all elements of $U, X \supset Y$ are true, it follows that either V, X are true. If it is V that is true, we are done. If it is X that is true, we know that Y is also true, meaning V would also be true by the assumptions.

N_1: We are given that at least one element of V is true whenever all elements of U, X are true. If U is true, either X is true or false: if it is true, the assumptions prove that V is also true. If it is false, then $\sim X$ would be true by definition.

N_2: We are given that when all elements of U are true, at least one element of V, X is true. Thus, if all elements of $U, \sim X$ are true, V would be the true element implied by U, as X is false.

Problem 5

(a): On the one hand, suppose that the sequent $U, X \to V$ is true under some valuation v. There are two cases: either at least one of U, X are false under v, or both of U, X are true under v. In the first case, if it is U that is false, $U \to V, \sim X$ is automatically true. Otherwise, if X is false, $\sim X$ is true, meaning $U \to V, \sim X$ is also true. On the other hand, if both of U, X are true under v, V is also true under v, therefore meaning that $U \to V, \sim X$ is also true under v.

On the other hand, suppose that $U \to V, \sim X$ is true under some valu-

ation v. If U is false, we are done, as this would trivially mean that $U, X \to V$ is true. On the other hand, if U is true, either V or $\sim X$ is true as well. If it is V, we are done, as $U, X \to V$ would be true. If it is $\sim X$, then X would be false, therefore meaning $U, X \to V$ is trivially true.

(b): On the one hand, suppose that $U \to V, X$ is true. If U is true, either V or X is true. If it is X that is true, then $U, \sim X \to V$ is trivially true. If it is V that is true, then $U, \sim X \to V$ is of course also true. If U is false under v, then $U, \sim X \to V$ is trivially true under v.

On the other hand, suppose that $U, \sim X \to V$ is true under v. If both of $U, \sim X$ are true, then X is false, therefore meaning $U \to V, X$ is trivially true. On the other hand, either of $U, \sim X$ would be false. If it is U that it false, $U \to X, V$ is trivially true. If it is $\sim X$ that is false, X is true, therefore meaning $U \to V, X$ is also true under v.

6

As for the axiom, suppose that $|S, TX, FX|$ is provable, where S is a set of signed formulas. Let $TX_1, ..., TX_n$ be all of the signed formulas with a T in S, and $FY_1, ..., FY_k$ be all of the signed formulas with an F in S. Thus $|S, TX, FX|$ is just $X_1, ..., X_n, X \to X, Y_1, ..., Y_k$, which is just the same thing as $U, X \to V, X$ holding, as we could just let $X_1, ..., X_n$ be the set U, and $Y_1, ..., Y_k$ be the set V. On the other hand, suppose that we take the axiom of the system G_0. Thus, for any sets U, V of formulas, and any formula X, $U, X \to X, V$ is provable. Again, consider some set S of signed formulas, and suppose that $TX_1, ..., TX_n$ be all of the signed formulas with a T in S, and $FY_1, ..., FY_k$ be all of the signed formulas with an F in S. Thus $|S, TX, FX|$ is just $X_1, ..., X_n, X \to X, Y_1, ..., Y_k$. Because the axiom of G_0 holds for any sets U and V, we can simply take U to be the set $X_1, ..., X_n$, and V to be the set $Y_1, ..., Y_k$, thus proving that $|S, TX, FX|$ is provable.

We now need to show that the inferences rules of G_0 in uniform notation are equivalent to those inferences rules of G_0 not in uniform notation. First, let us show that the rules in uniform notation imply those rules not in uniform notation.

C_1: Consider any two sets of formulas U and V. We can take S to be the set of formulas $TX_1, ..., TX_n, FY_1, ..., FY_k$ such that $X_1, ..., X_n = U$ and $Y_1, ..., Y_k = V$ (for making the following proofs shorter, we will call the set S constructed in this way the *signed set for* U, V. By A, if $|S, TX, TY|$ is provable, that is, $U, X, Y \to V$, so is $|S, TX \wedge Y|$, that is, $U, X \wedge Y \to V$.

C_2: Consider the signed set S for any two sets U and V. By B, if $|S, FX|$ is provable, that is, $U \to V, X$, and $|S, FY|$ is provable, that is, $U \to V, Y$, then $|S, FX \wedge Y|$ is provable, that is, $U \to V, X \wedge Y$.

D_1: Consider the signed set S for U and V. By A, if $|S, FX, FY|$ is provable, that is, $U \to V, X, Y$, then $|S, FX \vee Y|$ is also provable, that is, $U \to V, X \vee Y$.

D_2: Consider the signed set S for U and V. By B, if $|S, TX|$ is provable, that is, $U, X \to V$, and $|S, TY|$, that is, $U, Y \to V$, is provable, then so is $|S, TX \vee Y|$, that is, $U, X \vee Y \to V$.

I_1: Consider the signed set S for U and V. By A, if $|S, TX, FY|$, that is, $U, X \to V, Y$, is provable, so is $|S, FX \supset Y|$, that is, $U \to V, X \supset Y$.

I_2: Consider the signed set S for U and V. By B, if $|S, FX|$, that is $U \to V, X$ is provable, and $|S, TY|$, that is $U, Y \to V$, is provable, so is $|S, TX \supset Y|$, that is $U, X \supset Y \to V$.

N_1: Consider the signed set S for U and V. By A, if $|S, TX|$, that is $U, X \to V$, is provable, so is $|S, F{\sim}X|$, that is $U \to V, {\sim}X$.

N_2: Consider the signed set S for U and V. By A, if $|S, FX|$, that is $U \to V, X$, is provable, so is $|S, T{\sim}X|$, that is $U, {\sim}X \to V$.

We now turn to showing that the inferences rules of G_0 imply A and B.

A: If α is $TX \wedge Y$, this can be shown using C_1. If α is $FX \vee Y$, this can be shown using D_1. If α is $FX \supset Y$, this can be shown using I_1. If α is $T{\sim}X$, this can be shown using N_2. If α is $F{\sim}X$, this can be shown using N_1.

B: If β is $TX \vee Y$, this can be shown using D_2. If β is $FX \wedge Y$, this can be shown using C_2. If β is $TX \supset Y$, this can be shown using I_2. The cases where β is $T{\sim}X$ or $F{\sim}X$ are the same as in the α case.

A set S is unsatisfiable iff its corresponding sequent $|S|$ is a tautology

On the one hand, suppose that a non-empty set S of signed formulas is unsatisfiable. There are then multiple cases. In the first case, suppose that there are no formulas of the form TX. Thus, $|S|$ is of the form $\to Y_1, ..., Y_k$ where every Y_i is the formula FY_i without the sign F. We must show that at least one of $Y_1, ..., Y_k$ is true. We are given that $FY_1, ..., FY_k$ is unsatisfiable, meaning, under

every interpretation I, there is some formula FY_j that is false. Under this same interpretation I, it is obvious that Y_j (removing the sign) is true, thus meaning at least one of $FY_1, ..., FY_k$ is true. In the second case, suppose that the set S contains no formulas of the form FY. Thus $|S|$ is of the form $X_1, ...X_n \rightarrow$. We need to show that at least one of $TX_1, ..., TX_n$ is false. But this is immediate, as it was assumed that one of $TX_1, ..., TX_n$ was false under every interpretation I, which is the same as saying one of $X_1, ..., X_n$ is false under every interpretation I. In the last case, the set S contains at least one formula of the form TX and at least one of the form FY. Thus $|S|$ is of the form $X_1, ..., X_n \rightarrow Y_1, ..., Y_k$. If every element of $X_1, ..., X_n$ is true, this would mean, by the assumption that S is not simultaneously satisfiable, that at least one of $FY_1, ..., FY_k$ is false (call this element FY_K). As before, it can be easily seen that Y_k would then be true under this same interpretation, therefore meaning at least one of $Y_1, ..., Y_k$ is true.

On the other hand, suppose that $|S|$ is a tautology for some non-empty set of formulas S. There are then 3 cases:

Case 1: In this case, S does not contain any formulas of the form TX. We know that $\rightarrow Y_1, ..., Y_k$ is a tautology, meaning under every interpretation, at least one of $Y_1, ..., Y_k$, which is the same as saying, under every interpreation, at least one of $FY_1, ..., FY_k$ is false, thus meaning S is unsatisfiable.

Case 2: In this case, S does not contain any formulas of the form FY. We know that $X_1, ..., X_n \rightarrow$ is a tautology, meaning, under every interpretation, at least one of $X_1, ..., X_n$ are false, which is the same as saying at least one of $TX_1, ..., TX_n$ are false, which means that S is unsatisfiable.

Case 3: In this case, S contains at least one formula of the form FY, and at least one formula of the form TX. Thus we know that $X_1, ..., X_n \rightarrow Y_1, ..., Y_K$ is a tautology. There are then two cases. In the first case, at least one of $X_1, ..., X_n$ are false under an interpretation I, which is the same as saying at least one of $TX_1, ..., TX_n$ is false under I, which would mean S is not simultaneously true under I. In the other case, all of $X_1, ..., X_n$ are true, which would mean that at least one of $Y_1, ..., Y_k$ are true, which is the same as saying that at least one of $FY_1, ..., FY_k$ is false, therefore meaning that S is also not simultaneously satisfiable under this interpretation.

7

We first show that the altered tableau system is correct, that is, if a set of formulas S is satisfiable, then no altered tableau for S can close. It sufficies to show that, if S is satisfiable, then:

(1) If $A \in S$, then $S \bigcup A_1$ and $S \bigcup A_2$ are both satisfiable,
(2) If $B \in S$, then either $S \bigcup B_1$ or $S \bigcup B_2$ is satisfiable,
(3) If $C \in S$, then $S \bigcup C(a)$ is satisfiable for every parameter a,
(4) If $D \in S$, then $S \bigcup D(a)$ is satisfiable for every parameter a new to S.
These can all be proved by replacing $\alpha, \beta, \gamma, \delta$ by A, B, C, D, respectively, in Problem 2 of the Begginers Guide [Smullyan, 2014].

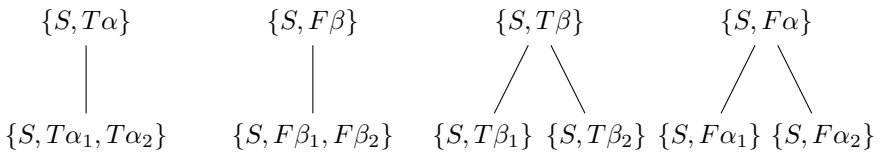
Next, we show that the altered tableau system is complete, that is, if an altered tableau for a set formulas S is open, then S is satisfiable. Consider a S set of the following form:
(1) If $A \in S$, then $A_1 \in S$ and $A_2 \in S$,
(2) If $B \in S$, then either $B_1 \in S$ or $B_2 \in S$,
(3) If $C \in S$, then $C(a) \in S$ for every parameter a,
(4) If $D \in S$, then $D(a) \in S$ for at least one parameter a.
(5) There is no formula $T\ X \in S$ such that $T \sim X \in S$; There is no formula X such that $T\ X \in S$ and $F\ X \in S$; and there is no formula X such that $F\ X \in S$ and $F \sim X \in S$.
Now, it is clear that if an altered tableau does not close, and runs on infinitely (i.e, is open), every branch on the altered tableau is such a set. Thus, if we can show that every set of the above form is satisfiable, we can show that, if an altered tableau for a set of formulas S is open, then the set S is simultaneously satisfiable (which would show by contraposition that the altered tableau system is complete). We can prove this by complete induction, that is, we must prove that there is some interpretation I under which all elements of S are true.

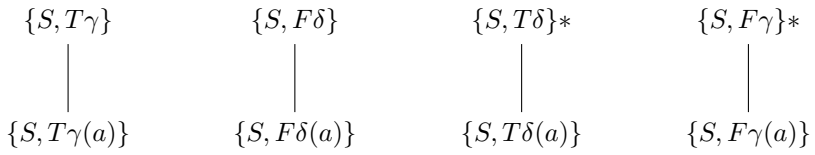
Base case: We must show that all formulas of degree 0 (atomic formulas) are satisfiable under some interpretation I (and then show all the formulas of greater degrees are true under that same interpretation). Now, we can assign the value T to any of the following formulas X: if $T\ X \in S$ or $F \sim X \in S$, then we assign X the value of truth (and hence both $T\ X$ and $F \sim X$ would be true as well). Otherwise, if $F\ X \in S$ or $T \sim X \in S$, then we assign X the value of falsity (and hence both $F\ X$ and $T \sim X$ would be true as well). If none of $T\ X, F \sim X, F\ X, T \sim X$ are in S, then it doesn't matter what value we assign to X.

Inductive case: Now, we assume that all formulas in S of a degree less than n are true under I. We must show that all formulas X of a degree $n+1$ in I are also true under I. Suppose X is of type A. Thus $A_1 \in S$ and $A_2 \in S$, which are thus both true under I, and hence A is also true under I. Next, suppose X is of type B. Thus either $B_1 \in S$ or $B_2 \in S$, and whichever one is, it is thus true under I, hence meaning B is also true under I. Next, suppose that X is of type C. Thus $C(a) \in S$ for every parameter a, which are all true under I, thus meaning C is also true under I. In the last case, suppose X is of type D. Thus $D(a) \in S$ for at least one parameter a, and these formulas are all true under I by the assumption, thus meaning D is also true under I. Hence, the proof is complete.

So, we have shown the above type of set to be satisfiable. This means that, because every open altered tableau is such a set, any set of formulas S which has an open altered tableau is satisfiable, and hence, by contraposition, the completeness of the altered tableau system has been proven.

Lastly, we must show that the system GG is complete, that is, we must show that for any sequent which is valid, GG provides a proof of that sequent. Now, let us transform any given sequent $X_1, ..., X_n \rightarrow Y_1, ..., Y_k$ to the set of formulas $\{T\ X_1, ..., T\ X_n, F\ Y_1, ..., F\ Y_k\}$. Now, if this set of formulas is unsatisfiable (which it must be if a sequent is valid), there is a proof of it using the altered tableau system (as the system was proven complete). So consider the proof of this set using the altered tableau system. First, let us convert the inference rules of the altered tableau method into their counterparts in uniform notation:





∗Providing a is not in the premise.